

Navigation Metrics Tables

Regrouped by world, method, and the three acquired trials

Input: /media/lorenzo/writable/Go2_navigation/bag_gp_tuning/metrics_from_bags/regrouped_trials

Output: /media/lorenzo/writable/Go2_navigation/bag_gp_tuning/metrics_from_bags/regrouped_trials/metrics_tables_report.pdf

World/Method Summary

World	Method	Trials	Valid bags	Inform.	Successes	SR3 %	SR inf %	T_goal succ mean s
indoor_office	baseline	3	3	2	1	33.33	50.00	84.82
indoor_office	bo_opti	3	3	2	1	33.33	50.00	52.26
open_world	baseline	3	2	2	1	33.33	50.00	41.55
open_world	bo_opti	3	2	2	2	66.67	100.0	41.94
warehouse	baseline	3	2	1	1	33.33	100.0	474.7
warehouse	bo_opti	3	3	2	1	33.33	50.00	188.5

World	Method	Path inf mean m	Final dist inf m	CE succ m	MPC mean ms	MPC p95 ms	MPC max ms	MPC samples
indoor_office	baseline	17.92	0.6079	0.0376	46.78	79.68	845.6	933
indoor_office	bo_opti	12.13	0.4984	0.0484	23.77	39.61	52.80	723
open_world	baseline	9.638	6.156	0.0590	40.85	59.06	109.8	1118
open_world	bo_opti	9.027	0.3165	0.0458	38.68	68.68	85.24	816
warehouse	baseline	73.97	0.4030	0.0271	60.78	111.2	1518.3	2529
warehouse	bo_opti	41.07	4.049	0.0573	48.41	77.38	1180.2	3922

World/Method Mean and Range

World	Method	Metric	Unit	Filter	N	Mean	Min	Max
indoor_office	baseline	Success	success flag	all_3_trials	3	0.3333	0	1.000
indoor_office	baseline	CE (m)	m	successful_runs	1	0.0376	0.0376	0.0376
indoor_office	baseline	Time-to-goal (s)	s	successful_informative_runs	1	84.82	84.82	84.82
indoor_office	baseline	Path length (m)	m	informative_runs	2	17.92	15.84	20.00
indoor_office	baseline	Final dist to goal (m)	m	informative_runs	2	0.6079	0.2670	0.9488
indoor_office	baseline	MPC solve mean (ms)	ms	informative_runs	2	46.78	39.11	54.46
indoor_office	baseline	MPC solve p95 (ms)	ms	informative_runs	2	79.68	66.98	92.38
indoor_office	baseline	MPC solve max (ms)	ms	informative_runs	2	845.6	765.6	925.5
indoor_office	baseline	MPC solve samples	samples	informative_runs	2	466.5	453.0	480.0
indoor_office	bo_opti	Success	success flag	all_3_trials	3	0.3333	0	1.000
indoor_office	bo_opti	CE (m)	m	successful_runs	1	0.0484	0.0484	0.0484
indoor_office	bo_opti	Time-to-goal (s)	s	successful_informative_runs	1	52.26	52.26	52.26
indoor_office	bo_opti	Path length (m)	m	informative_runs	2	12.13	11.66	12.60
indoor_office	bo_opti	Final dist to goal (m)	m	informative_runs	2	0.4984	0.3417	0.6551
indoor_office	bo_opti	MPC solve mean (ms)	ms	informative_runs	2	23.77	21.62	25.93
indoor_office	bo_opti	MPC solve p95 (ms)	ms	informative_runs	2	39.61	38.07	41.16
indoor_office	bo_opti	MPC solve max (ms)	ms	informative_runs	2	52.80	52.56	53.04
indoor_office	bo_opti	MPC solve samples	samples	informative_runs	2	361.5	324.0	399.0
open_world	baseline	Success	success flag	all_3_trials	3	0.3333	0	1.000
open_world	baseline	CE (m)	m	successful_runs	1	0.0590	0.0590	0.0590
open_world	baseline	Time-to-goal (s)	s	successful_informative_runs	1	41.55	41.55	41.55
open_world	baseline	Path length (m)	m	informative_runs	2	9.638	8.853	10.42
open_world	baseline	Final dist to goal (m)	m	informative_runs	2	6.156	0.3690	11.94
open_world	baseline	MPC solve mean (ms)	ms	informative_runs	2	40.85	39.60	42.10

World/Method Mean and Range

World	Method	Metric	Unit	Filter	N	Mean	Min	Max
open_world	baseline	MPC solve p95 (ms)	ms	informative_runs	2	59.06	58.84	59.28
open_world	baseline	MPC solve max (ms)	ms	informative_runs	2	109.8	75.50	144.0
open_world	baseline	MPC solve samples	samples	informative_runs	2	559.0	413.0	705.0
open_world	bo_opti	Success	success flag	all_3_trials	3	0.6667	0	1.000
open_world	bo_opti	CE (m)	m	successful_runs	2	0.0458	0.0403	0.0513
open_world	bo_opti	Time-to-goal (s)	s	successful_informative_runs	2	41.94	41.22	42.67
open_world	bo_opti	Path length (m)	m	informative_runs	2	9.027	8.984	9.069
open_world	bo_opti	Final dist to goal (m)	m	informative_runs	2	0.3165	0.2014	0.4317
open_world	bo_opti	MPC solve mean (ms)	ms	informative_runs	2	38.68	30.34	47.02
open_world	bo_opti	MPC solve p95 (ms)	ms	informative_runs	2	68.68	53.72	83.65
open_world	bo_opti	MPC solve max (ms)	ms	informative_runs	2	85.24	68.63	101.9
open_world	bo_opti	MPC solve samples	samples	informative_runs	2	408.0	361.0	455.0
warehouse	baseline	Success	success flag	all_3_trials	3	0.3333	0	1.000
warehouse	baseline	CE (m)	m	successful_runs	1	0.0271	0.0271	0.0271
warehouse	baseline	Time-to-goal (s)	s	successful_informative_runs	1	474.7	474.7	474.7
warehouse	baseline	Path length (m)	m	informative_runs	1	73.97	73.97	73.97
warehouse	baseline	Final dist to goal (m)	m	informative_runs	1	0.4030	0.4030	0.4030
warehouse	baseline	MPC solve mean (ms)	ms	informative_runs	1	60.78	60.78	60.78
warehouse	baseline	MPC solve p95 (ms)	ms	informative_runs	1	111.2	111.2	111.2
warehouse	baseline	MPC solve max (ms)	ms	informative_runs	1	1518.3	1518.3	1518.3
warehouse	baseline	MPC solve samples	samples	informative_runs	1	2529.0	2529.0	2529.0
warehouse	bo_opti	Success	success flag	all_3_trials	3	0.3333	0	1.000
warehouse	bo_opti	CE (m)	m	successful_runs	1	0.0573	0.0573	0.0573
warehouse	bo_opti	Time-to-goal (s)	s	successful_informative_runs	1	188.5	188.5	188.5

World/Method Mean and Range

World	Method	Metric	Unit	Filter	N	Mean	Min	Max
warehouse	bo_opti	Path length (m)	m	informative_runs	2	41.07	35.46	46.68
warehouse	bo_opti	Final dist to goal (m)	m	informative_runs	2	4.049	0.3094	7.788
warehouse	bo_opti	MPC solve mean (ms)	ms	informative_runs	2	48.41	28.80	68.01
warehouse	bo_opti	MPC solve p95 (ms)	ms	informative_runs	2	77.38	45.11	109.6
warehouse	bo_opti	MPC solve max (ms)	ms	informative_runs	2	1180.2	834.5	1525.9
warehouse	bo_opti	MPC solve samples	samples	informative_runs	2	1961.0	1673.0	2249.0

World/Method Mean and Range

World	Method	Range
indoor_office	baseline	1.000
indoor_office	baseline	0
indoor_office	baseline	0
indoor_office	baseline	4.161
indoor_office	baseline	0.6817
indoor_office	baseline	15.35
indoor_office	baseline	25.40
indoor_office	baseline	159.9
indoor_office	baseline	27.00
indoor_office	bo_opti	1.000
indoor_office	bo_opti	0
indoor_office	bo_opti	0
indoor_office	bo_opti	0.9358
indoor_office	bo_opti	0.3134
indoor_office	bo_opti	4.303
indoor_office	bo_opti	3.082
indoor_office	bo_opti	0.4857
indoor_office	bo_opti	75.00
open_world	baseline	1.000
open_world	baseline	0
open_world	baseline	0
open_world	baseline	1.570
open_world	baseline	11.57
open_world	baseline	2.497

World/Method Mean and Range

World	Method	Range
open_world	baseline	0.4399
open_world	baseline	68.50
open_world	baseline	292.0
open_world	bo_opti	1.000
open_world	bo_opti	0.0110
open_world	bo_opti	1.449
open_world	bo_opti	0.0843
open_world	bo_opti	0.2303
open_world	bo_opti	16.68
open_world	bo_opti	29.93
open_world	bo_opti	33.22
open_world	bo_opti	94.00
warehouse	baseline	1.000
warehouse	baseline	0
warehouse	baseline	0
warehouse	baseline	0
warehouse	baseline	0
warehouse	baseline	0
warehouse	baseline	0
warehouse	baseline	0
warehouse	baseline	0
warehouse	bo_opti	1.000
warehouse	bo_opti	0
warehouse	bo_opti	0

World	Method	Range
warehouse	bo_opti	11.22
warehouse	bo_opti	7.478
warehouse	bo_opti	39.21
warehouse	bo_opti	64.54
warehouse	bo_opti	691.4
warehouse	bo_opti	576.0

Baseline vs BO by Trial

World	Trial	baseline_success	bo_success	baseline_quality	bo_quality	baseline_Time-to-goal (s)	bo_Time-to-goal (s)	BO-base T_goal s
indoor_office	1	1	0	ok	ok	84.82		
indoor_office	2	0	1	ok	ok		52.26	
indoor_office	3	0	0	near_zero_path;low_path_ms	near_zero_path;low_path_ms			
open_world	1	0	1	ok	source_typo_opne_world		42.67	
open_world	2	1	1	ok	source_typo_opne_world	41.55	41.22	-0.3360
open_world	3	0	0	...	missing_trial			
warehouse	1	1	0	ok	ok	474.7		
warehouse	2	0	1	missing_trial	ok		188.5	
warehouse	3	0	0	near_zero_path;low_path_ms	near_zero_path;low_path_ms			

Baseline vs BO by Trial

World	Trial	baseline_Path length (m)	bo_Path length (m)	BO-base Path m	baseline_Final dist to goal (m)	bo_Final dist to goal (m)	BO-base Final dist to goal (m)	baseline_MPC solve mean (s)
indoor_office	1	15.84	12.60	-3.239	0.2670	0.6551	0.3880	39.11
indoor_office	2	20.00	11.66	-8.336	0.9488	0.3417	-0.6071	54.46
indoor_office	3	0.0000	0.0000	0.0000	8.976	8.976	0.0000	25.13
open_world	1	10.42	8.984	-1.439	11.94	0.2014	-11.74	42.10
open_world	2	8.853	9.069	0.2160	0.3690	0.4317	0.0627	39.60
open_world	3							
warehouse	1	73.97	46.68	-27.29	0.4030	7.788	7.385	60.78
warehouse	2		35.46			0.3094		
warehouse	3	0.0000	0.0000	-0.0000	8.976	8.976	-0.0000	35.89

Baseline vs BO by Trial

World	Trial	bo_MPC solve mean (ms)	BO-base MPC mean	baseline_MPC solve p95 (ms)	bo_MPC solve p95 (ms)	BO-base MPC p95	baseline_MPC solve max (ms)	bo_MPC solve max (ms)
indoor_office	1	21.62	-17.49	66.98	38.07	-28.91	765.6	53.04
indoor_office	2	25.93	-28.53	92.38	41.16	-51.22	925.5	52.56
indoor_office	3	18.69	-6.441	54.95	36.72	-18.22	57.81	47.09
open_world	1	30.34	-11.76	58.84	53.72	-5.126	144.0	68.63
open_world	2	47.02	7.421	59.28	83.65	24.36	75.50	101.9
open_world	3							
warehouse	1	68.01	7.228	111.2	109.6	-1.511	1518.3	1525.9
warehouse	2	28.80			45.11			834.5
warehouse	3	15.30	-20.59	45.34	34.63	-10.71	47.27	44.09

Baseline vs BO by Trial

World	Trial	BO-base MPC max
indoor_office	1	-712.6
indoor_office	2	-873.0
indoor_office	3	-10.72
open_world	1	-75.37
open_world	2	26.35
open_world	3	
warehouse	1	7.619
warehouse	2	
warehouse	3	-3.180

All Canonical Trials

World	Method	Trial	Success	SR_run (%)	CE (m)	T_goal s	Path m	Final dist m
indoor_office	baseline	1	1	100.0	0.0376	84.82	15.84	0.2670
indoor_office	baseline	2	0	0	0.0388		20.00	0.9488
indoor_office	baseline	3	0	0	0.0000		0.0000	8.976
indoor_office	bo_opti	1	0	0	0.0407		12.60	0.6551
indoor_office	bo_opti	2	1	100.0	0.0484	52.26	11.66	0.3417
indoor_office	bo_opti	3	0	0	0.0000		0.0000	8.976
open_world	baseline	1	0	0	0.0590		10.42	11.94
open_world	baseline	2	1	100.0	0.0590	41.55	8.853	0.3690
open_world	baseline	3	0	0				
open_world	bo_opti	1	1	100.0	0.0513	42.67	8.984	0.2014
open_world	bo_opti	2	1	100.0	0.0403	41.22	9.069	0.4317
open_world	bo_opti	3	0					
warehouse	baseline	1	1	100.0	0.0271	474.7	73.97	0.4030
warehouse	baseline	2	0					
warehouse	baseline	3	0	0	0.0000		0.0000	8.976
warehouse	bo_opti	1	0	0	0.0854		46.68	7.788
warehouse	bo_opti	2	1	100.0	0.0573	188.5	35.46	0.3094
warehouse	bo_opti	3	0	0	0.0000		0.0000	8.976

All Canonical Trials

World	Method	Trial	MPC mean ms	MPC p95 ms	MPC max ms	MPC samples	pose_source	goal_source
indoor_office	baseline	1	39.11	66.98	765.6	453.0	/go2/pose	goal_pose
indoor_office	baseline	2	54.46	92.38	925.5	480.0	/go2/pose	goal_pose
indoor_office	baseline	3	25.13	54.95	57.81	10.00	/go2/pose	path_endpoint
indoor_office	bo_opti	1	21.62	38.07	53.04	399.0	/go2/pose	path_endpoint
indoor_office	bo_opti	2	25.93	41.16	52.56	324.0	/go2/pose	path_endpoint
indoor_office	bo_opti	3	18.69	36.72	47.09	23.00	/go2/pose	path_endpoint
open_world	baseline	1	42.10	58.84	144.0	705.0	/go2/pose	goal_pose
open_world	baseline	2	39.60	59.28	75.50	413.0	/go2/pose	path_endpoint
open_world	baseline	3				0	missing	goal_pose
open_world	bo_opti	1	30.34	53.72	68.63	455.0	/go2/pose	path_endpoint
open_world	bo_opti	2	47.02	83.65	101.9	361.0	/go2/pose	path_endpoint
open_world	bo_opti	3						
warehouse	baseline	1	60.78	111.2	1518.3	2529.0	/go2/pose	goal_pose
warehouse	baseline	2						
warehouse	baseline	3	35.89	45.34	47.27	25.00	/go2/pose	path_endpoint
warehouse	bo_opti	1	68.01	109.6	1525.9	2249.0	/go2/pose	goal_pose
warehouse	bo_opti	2	28.80	45.11	834.5	1673.0	/go2/pose	goal_pose
warehouse	bo_opti	3	15.30	34.63	44.09	20.00	/go2/pose	path_endpoint

All Canonical Trials

World	Method	Trial	n_path_msgs	n_ref_points	valid_bag	near_zero_path	low_path_msgs	informative
indoor_office	baseline	1	82.00	740.0	1	0	0	1
indoor_office	baseline	2	103.0	1016.0	1	0	0	1
indoor_office	baseline	3	5.000	90.00	1	1	1	0
indoor_office	bo_opti	1	60.00	619.0	1	0	0	1
indoor_office	bo_opti	2	55.00	474.0	1	0	0	1
indoor_office	bo_opti	3	4.000	72.00	1	1	1	0
open_world	baseline	1	19.00	153.0	1	0	0	1
open_world	baseline	2	26.00	282.0	1	0	0	1
open_world	baseline	3	0	0	0	1	1	0
open_world	bo_opti	1	23.00	236.0	1	0	0	1
open_world	bo_opti	2	28.00	336.0	1	0	0	1
open_world	bo_opti	3			0			0
warehouse	baseline	1	371.0	3757.0	1	0	0	1
warehouse	baseline	2			0			0
warehouse	baseline	3	3.000	54.00	1	1	1	0
warehouse	bo_opti	1	77.00	844.0	1	0	0	1
warehouse	bo_opti	2	155.0	1710.0	1	0	0	1
warehouse	bo_opti	3	8.000	144.0	1	1	1	0

All Canonical Trials

World	Method	Trial	quality_flags
indoor_office	baseline	1	ok
indoor_office	baseline	2	ok
indoor_office	baseline	3	near_zero_path;low_path_msgs
indoor_office	bo_opti	1	ok
indoor_office	bo_opti	2	ok
indoor_office	bo_opti	3	near_zero_path;low_path_msgs
open_world	baseline	1	ok
open_world	baseline	2	ok
open_world	baseline	3	...
open_world	bo_opti	1	source_typo_opne_world
open_world	bo_opti	2	source_typo_opne_world
open_world	bo_opti	3	missing_trial
warehouse	baseline	1	ok
warehouse	baseline	2	missing_trial
warehouse	baseline	3	near_zero_path;low_path_msgs
warehouse	bo_opti	1	ok
warehouse	bo_opti	2	ok
warehouse	bo_opti	3	near_zero_path;low_path_msgs

Dropped Duplicate Rows

World	Method	Trial	Success	SR_run (%)	CE (m)	T_goal s	Path m	Final dist m
open_world	bo_opti	1	0	0				
open_world	bo_opti	2	0	0				

Dropped Duplicate Rows

World	Method	Trial	MPC mean ms	MPC p95 ms	MPC max ms	MPC samples	pose_source	goal_source
open_world	bo_opti	1				0	missing	goal_pose
open_world	bo_opti	2				0	missing	goal_pose

Dropped Duplicate Rows

World	Method	Trial	n_path_msgs	n_ref_points	valid_bag	near_zero_path	low_path_msgs	informative
open_world	bo_opti	1	0	0	0	1	1	0
open_world	bo_opti	2	0	0	0	1	1	0

Dropped Duplicate Rows

World	Method	Trial	quality_flags
open_world	bo_opti	1	...
open_world	bo_opti	2	...

World	Method	Trial	Success	SR_run (%)	CE (m)	T_goal s	Path m	Final dist m
indoor_office	baseline	1	1	100.0	0.0376	84.82	15.84	0.2670
indoor_office	bo_opti	1	0	0	0.0407		12.60	0.6551
indoor_office	baseline	2	0	0	0.0388		20.00	0.9488
indoor_office	bo_opti	2	1	100.0	0.0484	52.26	11.66	0.3417
indoor_office	baseline	3	0	0	0.0000		0.0000	8.976
indoor_office	bo_opti	3	0	0	0.0000		0.0000	8.976

World	Method	Trial	MPC mean ms	MPC p95 ms	MPC max ms	MPC samples	pose_source	goal_source
indoor_office	baseline	1	39.11	66.98	765.6	453.0	/go2/pose	goal_pose
indoor_office	bo_opti	1	21.62	38.07	53.04	399.0	/go2/pose	path_endpoint
indoor_office	baseline	2	54.46	92.38	925.5	480.0	/go2/pose	goal_pose
indoor_office	bo_opti	2	25.93	41.16	52.56	324.0	/go2/pose	path_endpoint
indoor_office	baseline	3	25.13	54.95	57.81	10.00	/go2/pose	path_endpoint
indoor_office	bo_opti	3	18.69	36.72	47.09	23.00	/go2/pose	path_endpoint

World	Method	Trial	n_path_msgs	n_ref_points	valid_bag	near_zero_path	low_path_msgs	informative
indoor_office	baseline	1	82.00	740.0	1	0	0	1
indoor_office	bo_opti	1	60.00	619.0	1	0	0	1
indoor_office	baseline	2	103.0	1016.0	1	0	0	1
indoor_office	bo_opti	2	55.00	474.0	1	0	0	1
indoor_office	baseline	3	5.000	90.00	1	1	1	0
indoor_office	bo_opti	3	4.000	72.00	1	1	1	0

World	Method	Trial	quality_flags
indoor_office	baseline	1	ok
indoor_office	bo_opti	1	ok
indoor_office	baseline	2	ok
indoor_office	bo_opti	2	ok
indoor_office	baseline	3	near_zero_path;low_path_msgs
indoor_office	bo_opti	3	near_zero_path;low_path_msgs

World	Method	Trial	Success	SR_run (%)	CE (m)	T_goal s	Path m	Final dist m
open_world	baseline	1	0	0	0.0590		10.42	11.94
open_world	bo_opti	1	1	100.0	0.0513	42.67	8.984	0.2014
open_world	baseline	2	1	100.0	0.0590	41.55	8.853	0.3690
open_world	bo_opti	2	1	100.0	0.0403	41.22	9.069	0.4317
open_world	baseline	3	0	0				
open_world	bo_opti	3	0					

World	Method	Trial	MPC mean ms	MPC p95 ms	MPC max ms	MPC samples	pose_source	goal_source
open_world	baseline	1	42.10	58.84	144.0	705.0	/go2/pose	goal_pose
open_world	bo_opti	1	30.34	53.72	68.63	455.0	/go2/pose	path_endpoint
open_world	baseline	2	39.60	59.28	75.50	413.0	/go2/pose	path_endpoint
open_world	bo_opti	2	47.02	83.65	101.9	361.0	/go2/pose	path_endpoint
open_world	baseline	3				0	missing	goal_pose
open_world	bo_opti	3						

World	Method	Trial	n_path_msgs	n_ref_points	valid_bag	near_zero_path	low_path_msgs	informative
open_world	baseline	1	19.00	153.0	1	0	0	1
open_world	bo_opti	1	23.00	236.0	1	0	0	1
open_world	baseline	2	26.00	282.0	1	0	0	1
open_world	bo_opti	2	28.00	336.0	1	0	0	1
open_world	baseline	3	0	0	0	1	1	0
open_world	bo_opti	3			0			0

World	Method	Trial	quality_flags
open_world	baseline	1	ok
open_world	bo_opti	1	source_typo_opne_world
open_world	baseline	2	ok
open_world	bo_opti	2	source_typo_opne_world
open_world	baseline	3	...
open_world	bo_opti	3	missing_trial

Warehouse Trials

World	Method	Trial	Success	SR_run (%)	CE (m)	T_goal s	Path m	Final dist m
warehouse	baseline	1	1	100.0	0.0271	474.7	73.97	0.4030
warehouse	bo_opti	1	0	0	0.0854		46.68	7.788
warehouse	baseline	2	0					
warehouse	bo_opti	2	1	100.0	0.0573	188.5	35.46	0.3094
warehouse	baseline	3	0	0	0.0000		0.0000	8.976
warehouse	bo_opti	3	0	0	0.0000		0.0000	8.976

World	Method	Trial	MPC mean ms	MPC p95 ms	MPC max ms	MPC samples	pose_source	goal_source
warehouse	baseline	1	60.78	111.2	1518.3	2529.0	/go2/pose	goal_pose
warehouse	bo_opti	1	68.01	109.6	1525.9	2249.0	/go2/pose	goal_pose
warehouse	baseline	2						
warehouse	bo_opti	2	28.80	45.11	834.5	1673.0	/go2/pose	goal_pose
warehouse	baseline	3	35.89	45.34	47.27	25.00	/go2/pose	path_endpoint
warehouse	bo_opti	3	15.30	34.63	44.09	20.00	/go2/pose	path_endpoint

World	Method	Trial	n_path_msgs	n_ref_points	valid_bag	near_zero_path	low_path_msgs	informative
warehouse	baseline	1	371.0	3757.0	1	0	0	1
warehouse	bo_opti	1	77.00	844.0	1	0	0	1
warehouse	baseline	2			0			0
warehouse	bo_opti	2	155.0	1710.0	1	0	0	1
warehouse	baseline	3	3.000	54.00	1	1	1	0
warehouse	bo_opti	3	8.000	144.0	1	1	1	0

World	Method	Trial	quality_flags
warehouse	baseline	1	ok
warehouse	bo_opti	1	ok
warehouse	baseline	2	missing_trial
warehouse	bo_opti	2	ok
warehouse	baseline	3	near_zero_path;low_path_msgs
warehouse	bo_opti	3	near_zero_path;low_path_msgs